
DNV-RP-C203 · API RP 2A-WSD · BS 7608 · EN 1993-1-9 · NORSOK N-004 · BS 7910

Fatigue & fracture — S-N life and crack growth

A code-referenced fatigue toolkit: a 221-curve S-N library spanning DNV-RP-C203, API RP 2A-WSD, BS 7608, Eurocode 3 and NORSOK N-004; Efthymiou stress-concentration factors for tubular joints; rainflow counting into Miner's-rule damage; hotspot-stress extrapolation; and Paris'-law crack growth. Every curve, SCF equation and read-out formula carries its clause reference in-code, with numeric checks pinned to published tables and hand calculations.

221

S-N curves, 6 standards

17

standard curve sets

3

environments (air / seawater+CP / free corrosion)

WHAT YOU GET

- Single S-N lookup across 17 standard sets keyed by curve id (e.g. DNV-RP-C203:D:air) with slope, knee and thickness-correction metadata
- Efthymiou (OTC 4829) SCFs for T/Y and K tubular joints under axial, in-plane and out-of-plane bending, plus DNV plate / weld SCFs
- Weld-fatigue quick-check: stress-range histogram + weld class → thickness correction → Palmgren-Miner damage, pass/fail and life factor
- Two- and three-point hotspot-stress extrapolation to the weld toe (DNV-RP-C203 §4.3 · IIW)
- Paris'-law crack-growth integration (BS 7910) from initial to critical crack size — the fracture-mechanics complement to the S-N route